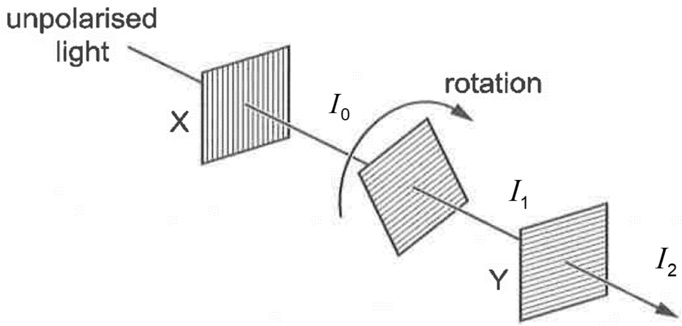
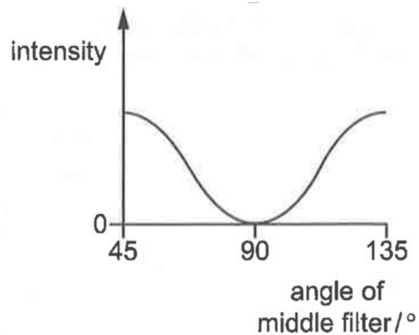
		$= 1.0\pi$ rad (expressed to within 2π rad)
17	**B**	By Malus' law which states that the intensity of a beam of plane-polarised light after passing through a rotatable polariser varies as the square of the cosine of the angle through which the polariser is rotated from the position that gives maximum intensity.  $I_1 = I_0 \cos^2\theta$ $I_2 = I_1 \cos^2(90^\circ - \theta)$ $= I_0 \cos^2\theta \cos^2(90^\circ - \theta)$ For $45^\circ \le \theta \le 135^\circ$, the variation of intensity of light emerging from filter Y with angle of middle filter θ is as following:

No.	Answer	Solution
		 By trial and error, $\theta = 45^\circ, I_2 = I_0 \cos^2 45^\circ \cos^2\left(90^\circ - 45^\circ\right)$ $= 0.25 I_0$ $\theta = 90^\circ, I_2 = I_0 \cos^2 90^\circ \cos^2\left(90^\circ - 90^\circ\right)$ $= 0$ $\theta = 135^\circ, I_2 = I_0 \cos^2 135^\circ \cos^2\left(90^\circ - 135^\circ\right)$ $= 0.25 I_0$
18	D	